

NG
S
FER

AEGIS

HEXAPOD ROBOT

INNOVATION | MOBILITY | SAFER TOMORROWS



MARKET
REAL DEMAND



ECONOMY
JOBS. GROWTH



SOUTH AFRICA
LOCAL IMPACT



CAL SOLUTIONS
AL IMPACT

HEXAPOD ROBOT

AEGIS

Hexapod Robot

Introduction

A hexapod robotic system is an automated mechanism that utilises multiple legs, typically six, for locomotive purposes. The hexapedal design provides a stable platform, enabling versatile traversal across various types of terrain [1]. Furthermore, its unique kinematic structure allows the robot to climb obstacles and manoeuvre through highly constrained spaces, such as narrow cracks and fissures [1].

By emulating the structural characteristics of a spider, the hexapedal design provides flexibility and replicability [2] necessary for the robot to adapt to diverse operational environments. Due to these locomotive advantages, hexapod robots are widely utilised in industrial, military, aerospace, and emergency rescue fields. These demanding sectors frequently require operation in hazardous and unstable environments.

The primary focus of this project is to develop a biomimetic robotic mechanism tailored for emergency rescue operations by evaluating existing systems, generating alternative conceptual designs, and selecting an optimal design for manufacturing.

Aim

To design, simulate, and construct an autonomous hexapod robot capable of traversing highly unstable, unstructured terrain to accelerate survivor localization in post-disaster environments.

Problem Statement

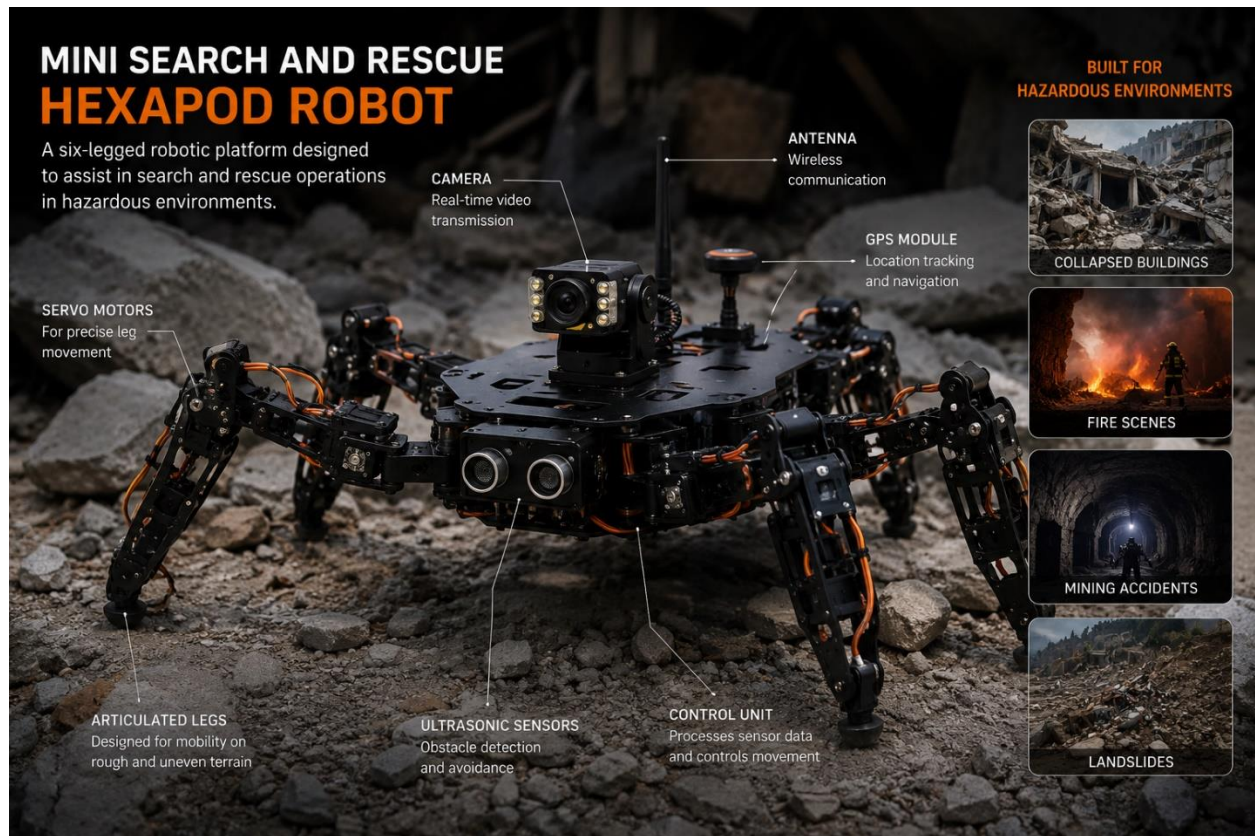
Following seismic events or similar natural disasters, structural collapses frequently result in the entrapment of individuals within the debris field. Rescue teams face critical challenges when trying to locate survivors efficiently due to the non-uniform and unstable nature of the collapsed material. Consequently, the traditional search and rescue approach often results in severe delays while simultaneously exposing rescue personnel to hazardous conditions.

A design team from the University of Johannesburg is developing a hexapod robotic system. This robotic platform aims to mitigate risks to personnel by penetrating high-density debris fields to locate trapped individuals with increased speed and precision.

Proposed Solution

To address the challenges faced during emergency search-and-rescue operations, our team proposes the development of a Mini Search and Rescue Hexapod Robot, a six-legged robotic platform specifically designed to assist emergency response teams in hazardous and inaccessible environments. Rather than replacing human rescuers, the robot is intended to serve as an intelligent support system by entering dangerous areas first, collecting critical information, and providing rescue teams with a clearer understanding of the situation before

they begin rescue operations. The robot will also be equipped with sensors and a vision system capable of collecting real-time environmental information. These technologies will allow the robot to transmit live video and other critical data to emergency responders, enabling them to assess structural conditions, identify obstacles, and locate potential survivors while remaining at a safe distance.



- A six-legged platform designed for disaster environments.
- Intelligent motion using inverse kinematics.
- Real-time environmental sensing.
- A modular architecture that can be upgraded with AI and additional sensors.
- A focus on supporting emergency responders rather than replacing them.

The project integrates mechanical engineering, electronics, computer science, and artificial intelligence to develop a mobile robotic system capable of supporting rescue personnel by exploring disaster-affected areas and providing real-time environmental information.

- Lack of information is the main problem.
- Time is wasted searching the wrong places.

- Victims' survival chances decrease.
- Rescuers are placed at unnecessary risk.
- Therefore, a better solution is needed.

Our vision is simple: no emergency responder should have to risk a life to discover whether a path is safe, and no victim should lose the opportunity for rescue because critical information could not be obtained in time.

Why Six Legs?

A six-legged configuration provides an important advantage for search-and-rescue applications:

Stability

With six legs, the robot can maintain multiple points of contact with the ground.

Terrain Adaptability

Individual legs can be positioned to deal with uneven surfaces and obstacles.

Redundancy

If one leg temporarily loses contact with the ground, the remaining legs can still provide support.

Compact Mobility

Unlike wheeled robots, the robot does not require a continuous smooth surface.

Marketing And Economic

From Student Prototype → South African Robotics Solution

The technology has potential beyond a university project.

Potential Markets

Sector	Possible Application
Mining	Remote inspection of hazardous areas
Emergency Services	Search and reconnaissance
Construction	Inspection of unstable structures
Energy	Infrastructure inspection

Industry	Remote equipment inspection
Disaster Management	Post-collapse reconnaissance

Long-Term Vision

**Start with an affordable prototype → validate the technology → develop a stronger platform
→ target industrial and emergency-response applications.**

Buildability

1. Mechanical System

- Six legs
- Coxa, femur and tibia
- Robot body
- Servo mounts
- 3D-printed components

2. Electronics

- ESP32
- Servo drivers
- Battery
- Voltage regulators
- Wiring

3. Sensing

- Camera
- Ultrasonic / ToF sensors
- LiDAR
- IMU

4. Software

- Inverse kinematics

- Gait generation
- Servo control
- Wireless communication
- Simulation

5. Testing

Simulation → Individual leg → Full robot → Controlled environment → Field testing

This reduces development risk and allows the team to validate the system before investing in a complete physical robot.

Estimated prototype cost

Estimated Hardware BOM — South African Prices

For a practical 18-servo hexapod (3 servos × 6 legs), the following is a realistic planning range.

Estimated Hardware BOM — South African Prices

For a practical 18-servo hexapod (3 servos × 6 legs), the following is a realistic planning range.

Component	Quantity	Estimated Unit Price	Estimated Total
MG996R metal-gear servo	18	R95–R250	R1,710–R4,500
ESP32 development board	1	R98–R225	R98–R225
PCA9685 16-channel servo driver	2	R98–R150	R196–R300
ESP32 camera / camera module	1	R250–R450	R250–R450
Ultrasonic sensors	2	R25–R50	R50–R100
IMU sensor	1	R50–R150	R50–R150
LiDAR / ToF sensor	1	R300–R700	R300–R700
Battery + holder/power system	1	R200–R600	R200–R600
5V buck regulator	1	R70–R225	R70–R225
3D-printed mechanical structure	1 set	R250–R500	R250–R500
Wires, connectors, switches	1 set	R150–R300	R150–R300

Fasteners/bearings/misc. hardware	1 set	R200–R400	R200–R400
TOTAL ESTIMATE			≈ R3,524–R8,450

Cost Strategy

1. Start Small

Build and test one robotic leg first.

2. Simulate Before Buying

Use simulation to validate:

- Inverse kinematics
- Joint movement
- Gait patterns
- Walking stability

3. Use Commercially Available Components

ESP32, MG996R servos and PCA9685 controllers are relatively accessible components in the South African electronics market.

4. 3D Printing

Instead of machining every mechanical component, the body and leg components can be manufactured using 3D printing.

A 1 kg PLA spool is currently around R229–R278 from a South African robotics supplier, providing a relatively low-cost route for prototype parts.

5. Modular Design

If one component fails:

Replace the module — not the entire robot.

This project is being developed by a small team of three undergraduate students: a second-year Computer Science.

major, a first-year Mathematics and Computer Science major, and a third-year Mechanical Engineering major. The Mechanical Engineering student will lead the structural design, mechanical assembly, and physical integration of components. The Computer Science student will focus on software development, integrating sensors, cameras, control systems, and data processing. The Mathematics and Computer Science student will handle algorithmic aspects, including leg movement kinematics, path memorization, mapping, and optimization.



Phase 1: Research & Mechanical Design 5

Fatima leads the creation of the physical structure, leg configuration, materials, grip mechanisms, self-leveling system, and initial CAD/sketches. Team conducts joint research on components. Milestone: Approved mechanical blueprint and parts list. Initial prototype frame or single-leg test.

Phase 2: Mathematical Modeling (Overlapping with Phase 1)



Awande Zubane develops the core math: leg kinematics, balance/self-leveling algorithms, path memorization, surface adaptation, and basic optimization. Milestone: Documented mathematical models and simulations.

Phase 3: Software Development & Integration



Noor Ain implements control software, sensor fusion, live feed, manual control interface, path recording/return functionality, and communication systems. Milestone: Working software simulation and basic hardware-in-the-loop testing.

References

- [1] <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9087504>
- [2] <https://iopscience.iop.org/article/10.1088/1742-6596/1168/2/022094/pdf>
- [3] <https://journal.umy.ac.id/index.php/jrc/article/view/23905/11435>
<https://www.sciencedirect.com/science/article/pii/S1364032109000318>
https://www.researchgate.net/profile/Kenneth-Ibekwe2/publication/377897752_Renewable_energy_projects_in_Africa_A_review_of_climate_finance_strategies/links/65cbc6ca34bbff5ba70ef93b/Renewable-energy-projects-in-Africa-A-reviewof-climate-finance-strategies.pdf